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Binary Change Detection on EO-SAR Image Pairs using a Lightweight Early-Fusion U-Net

Technical Assessment Submission for GalaxEye Space

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Executive Summary

This work addresses the problem of binary change detection using co-registered Electro-Optical (EO) and Synthetic Aperture Radar (SAR) image pairs.The objective was to predict binary pixel-level change masks between pre-event EO imagery and post-event SAR imagery. Compared to conventional EO-EO change detection, the EO-SAR setting is considerably more difficult due to the large modality gap, SAR speckle noise, and substantial appearance differences between the two sensing modalities.

Initial exploratory analysis showed several dataset-specific challenges. Positive change pixels occupied only a very small fraction of the total image area in most scenes, leading to severe class imbalance. In addition, many samples contained invalid or partially overlapping regions near image boundaries, which introduced strong artificial edges and unreliable supervision. SAR intensity distributions were also found to vary noticeably between training and test splits, indicating a moderate train-test distribution shift.

To address these issues, a lightweight early-fusion U-Net architecture was implemented using EO RGB channels concatenated with normalized SAR input. The training pipeline incorporated masked focal and Dice losses, validity masking, and positive-biased patch sampling to improve learning on sparse change regions. During inference, sliding-window prediction with Gaussian blending and geometric test-time augmentation (TTA) was applied. Final predictions were refined using connected-component filtering and invalid-region suppression.

On the provided test split, the final model achieved an IoU of 0.1417 and an F1-score of 0.2482. The corresponding Precision, Recall, and MCC values were 0.1735, 0.4356, and 0.2400 respectively. Validation performance was substantially higher than test performance, suggesting limited generalization under the observed distribution shift. Qualitative analysis showed that many false positives originated from SAR texture variations, urban edges, and invalid border regions. Despite these limitations, the final pipeline demonstrates a practical and computationally lightweight approach for EO-SAR change detection under constrained training conditions.

1. Introduction

1.1 Remote Sensing Change Detection

Remote sensing change detection involves identifying meaningful differences between observations of the same geographic region acquired at different times. In satellite imagery, this is commonly used to monitor urban development, environmental changes, flooding, wildfires, and post-disaster damage. The task is typically formulated as a pixel-wise prediction problem where the model determines whether a region has changed or remained unchanged between two acquisitions.

In disaster response scenarios, rapid change detection can help estimate damaged infrastructure, identify heavily affected regions, and support resource allocation during emergency operations. Since satellite imagery can cover large geographic areas quickly, automated change detection systems are useful for generating damage assessments at a much larger scale than manual inspection.

1.2 EO and SAR Satellite Imagery

This project focuses on multimodal change detection using paired Electro-Optical (EO) and Synthetic Aperture Radar (SAR) imagery. EO imagery captures reflected visible light and therefore contains semantic appearance information such as color, texture, roads, vegetation, and building structures. These images are generally easier for humans to interpret visually.

SAR imagery is fundamentally different. Instead of visible light, SAR sensors measure microwave backscatter responses from the Earth's surface. This allows SAR systems to operate even under cloud cover or poor lighting conditions, which is particularly useful during natural disasters where optical imagery may be unavailable or partially obstructed. However, SAR imagery also contains strong speckle noise and geometric distortions that make interpretation more difficult compared to EO imagery.

1.3 Challenges in EO-SAR Change Detection

EO-SAR change detection is considerably more difficult than conventional EO-EO settings because the two modalities represent the same scene in very different ways. A structure that appears visually clear in EO imagery may produce complex or noisy scattering patterns in SAR imagery. As a result, direct appearance matching between the two modalities is unreliable.

Another practical challenge is that co-registered EO and SAR images are not always perfectly aligned. Small registration inconsistencies can produce artificial boundaries or noisy transitions near image edges. During exploratory analysis, several samples were also observed to contain partially invalid overlap regions, particularly near scene borders. These regions can introduce false activations if they are not handled carefully during training and evaluation.

1.4 Class Imbalance and Sparse Change Regions

A major difficulty in disaster-related change detection datasets is severe class imbalance. In most scenes, unchanged regions occupy the vast majority of pixels, while actual damage regions are relatively sparse. This creates a strong bias toward predicting the background class.

Under these conditions, standard accuracy metrics can become misleading. A model may achieve very high accuracy simply by predicting most pixels as unchanged, while still failing to detect meaningful damaged regions. Sparse positives also make optimization more unstable, especially during early training stages. Because of this, handling imbalance became an important part of the overall pipeline design.

1.5 Objective of This Work

The objective of this work was to build a practical EO-SAR binary change detection pipeline capable of identifying damaged regions from paired pre-event EO and post-event SAR imagery. The implementation focused on maintaining a relatively lightweight and interpretable system while addressing dataset-specific challenges such as sparse positives, SAR noise, invalid overlap regions, and distribution shift.

Rather than implementing large multimodal transformer architectures, the project primarily focused on preprocessing quality, stable training behavior, masking strategies, threshold analysis, and postprocessing refinements. The later sections discuss the exploratory analysis, methodology, experiments, and observed limitations of the final pipeline in detail.

2. Problem Statement

Binary change detection is the task of identifying regions that have changed between two observations of the same geographic area acquired at different times. In this assignment, the input consists of paired Electro-Optical (EO) and Synthetic Aperture Radar (SAR) imagery collected before and after a disaster event. The objective is to predict a binary pixel-wise mask indicating whether each location belongs to the change or no-change class.

The provided dataset contains co-registered EO-SAR image pairs along with pixel-level annotations. Following the assignment specification, the original four semantic classes were remapped into a binary setting where background and intact regions were assigned to the no-change class, while damaged and destroyed regions were assigned to the change class.

Compared to standard EO-EO change detection, the EO-SAR setting introduces several additional difficulties. EO imagery contains texture, color, and semantic appearance information that is relatively intuitive to interpret visually. In contrast, SAR imagery represents microwave backscatter responses and is strongly affected by surface geometry, material properties, and speckle noise. As a result, the same structure can appear very differently across the two modalities even in the absence of real change.

Another major challenge in the dataset is severe class imbalance. In many scenes, change pixels occupy only a very small percentage of the total image area, while no-change pixels dominate the distribution. This makes conventional accuracy metrics unreliable since a model can achieve high accuracy while still failing to detect meaningful changes. Additionally, several samples contain invalid or partially overlapping regions near image boundaries due to imperfect co-registration between EO and SAR acquisitions. These regions can introduce artificial edges and produce false activations during inference if not handled properly.

The goal of this work was therefore not only to train a segmentation model, but also to design a practical pipeline that could handle sparse positives, noisy SAR inputs, invalid regions, and moderate train-test distribution shift while maintaining reasonable computational efficiency.

3. Literature Survey

Deep learning based change detection has become increasingly common in remote sensing, especially for disaster assessment and urban monitoring tasks. Most recent methods treat change detection as a semantic segmentation problem, where the model predicts a binary or multi-class mask from paired observations acquired at different times. However, EO-SAR change detection remains more difficult than conventional EO-EO settings because the two modalities capture fundamentally different physical properties of the scene. Before implementing the final pipeline, several papers were reviewed to better understand common architectural choices, preprocessing strategies, and dataset-specific challenges such as class imbalance, SAR noise, and cross-modal feature mismatch.

Paper	**Core Idea**	**What Was Learned**	**Relevance To This Work**
U-Net: Convolutional Networks for Biomedical Image Segmentation (2015)	Encoder-decoder segmentation architecture with skip connections for spatial localization	U-Net remains a strong baseline for dense prediction tasks with limited training data and relatively simple training requirements	Directly influenced the final architecture choice. A lightweight U-Net was selected because it was easier to train and debug under limited compute constraints
Fully Convolutional Siamese Networks for Change Detection (2018)	Early-fusion change detection by concatenating multi-temporal inputs before a shared encoder	Early fusion can work surprisingly well for change detection without requiring complicated dual-stream architectures	Direct inspiration for the EO + SAR concatenation strategy used in this implementation
Deep Learning for Change Detection in Remote Sensing Images: Comprehensive Review and Meta-Analysis (2020)	Survey covering traditional and deep learning based change detection methods	Helped identify common problems in remote sensing change detection including imbalance, registration errors, and evaluation challenges	Used mainly for baseline understanding of the field and evaluation methodology

Paper	Core Idea	What Was Learned	Relevance To This Work
SpaceNet 6: Multi-Sensor All Weather Mapping Dataset (2020)	Building extraction and change-related analysis using SAR imagery in challenging conditions	Highlighted practical difficulties of SAR imagery including speckle noise, geometric distortions, and distribution variability	Influenced preprocessing decisions, SAR normalization experiments, and qualitative failure analysis

The U-Net architecture had the strongest influence on the final implementation. The primary objective of this assignment was not to develop a very large multimodal network, but to build a stable and interpretable pipeline under limited compute and time constraints. A lightweight encoder-decoder structure was therefore preferred over transformer-based or attention-heavy approaches. The skip connections used in U-Net were particularly useful for recovering spatial details in sparse change regions during decoding.

The FC-EF baseline demonstrated that relatively simple early-fusion pipelines can already provide competitive change detection performance. This aligned well with the practical constraints of the assignment. Instead of designing separate EO and SAR branches with complex fusion stages, EO RGB channels and SAR inputs were concatenated directly before entering the network. This simplified experimentation, reduced training complexity, and made debugging easier during early development.

The survey literature was mainly useful for understanding common failure modes in remote sensing change detection. Many of the issues discussed in prior work also appeared during experimentation, including severe foreground sparsity, threshold sensitivity, noisy predictions, and challenges caused by imperfect co-registration. This influenced the decision to include threshold analysis, MCC evaluation, and detailed qualitative error inspection rather than relying only on aggregate metrics such as accuracy.

The SpaceNet 6 work and related SAR-focused studies were useful for understanding how noisy SAR imagery behaves in segmentation pipelines. During experiments, many false positives appeared around textured SAR regions, urban edges, and invalid border areas. Reviewing SAR-focused literature helped contextualize these behaviors and motivated additional preprocessing and postprocessing steps such as per-image SAR normalization, Gaussian sliding-window blending, connected-component filtering, and invalid-region suppression.

More recent multimodal fusion approaches based on transformers and attention mechanisms were also briefly reviewed during the initial planning phase. However, these methods were not implemented in the final pipeline because they would have substantially increased training complexity and compute requirements. They are instead considered as possible future work directions.

4. Dataset Description and Exploratory Data Analysis

4.1 Dataset Overview

The dataset provided for the assignment consists of co-registered pre-event and post-event EO-SAR image pairs along with pixel-level annotation masks. The task is binary change detection, where each pixel must be classified as either change or no-change. Following the assignment instructions, the original four semantic classes were remapped into two categories by grouping damaged and destroyed regions as the positive change class, while background and intact regions were treated as no-change.

The dataset was provided with fixed train, validation, and test splits. The test split available during development represented only a partial subset of the final hidden evaluation data. This made generalization an important consideration during model development rather than optimizing only for the visible test samples.

The input data contains multimodal imagery where the pre-event image is EO RGB imagery and the post-event image is SAR imagery. Compared to standard optical change detection datasets, the EO-SAR setup introduces larger appearance differences between the two temporal observations.

4.2 Visual Inspection of Samples

Initial visual inspection of the dataset was performed before implementing the training pipeline. Several representative training samples are shown in Figure 1.

A few observations became immediately noticeable during inspection:

- Positive change regions were often sparse and localized.
- Some scenes contained almost no positive pixels.
- SAR imagery showed strong texture variability across different scenes.
- Certain samples contained partially invalid or weakly aligned border regions.
- Urban regions produced very different visual patterns between EO and SAR modalities.

These observations influenced later preprocessing and sampling decisions.

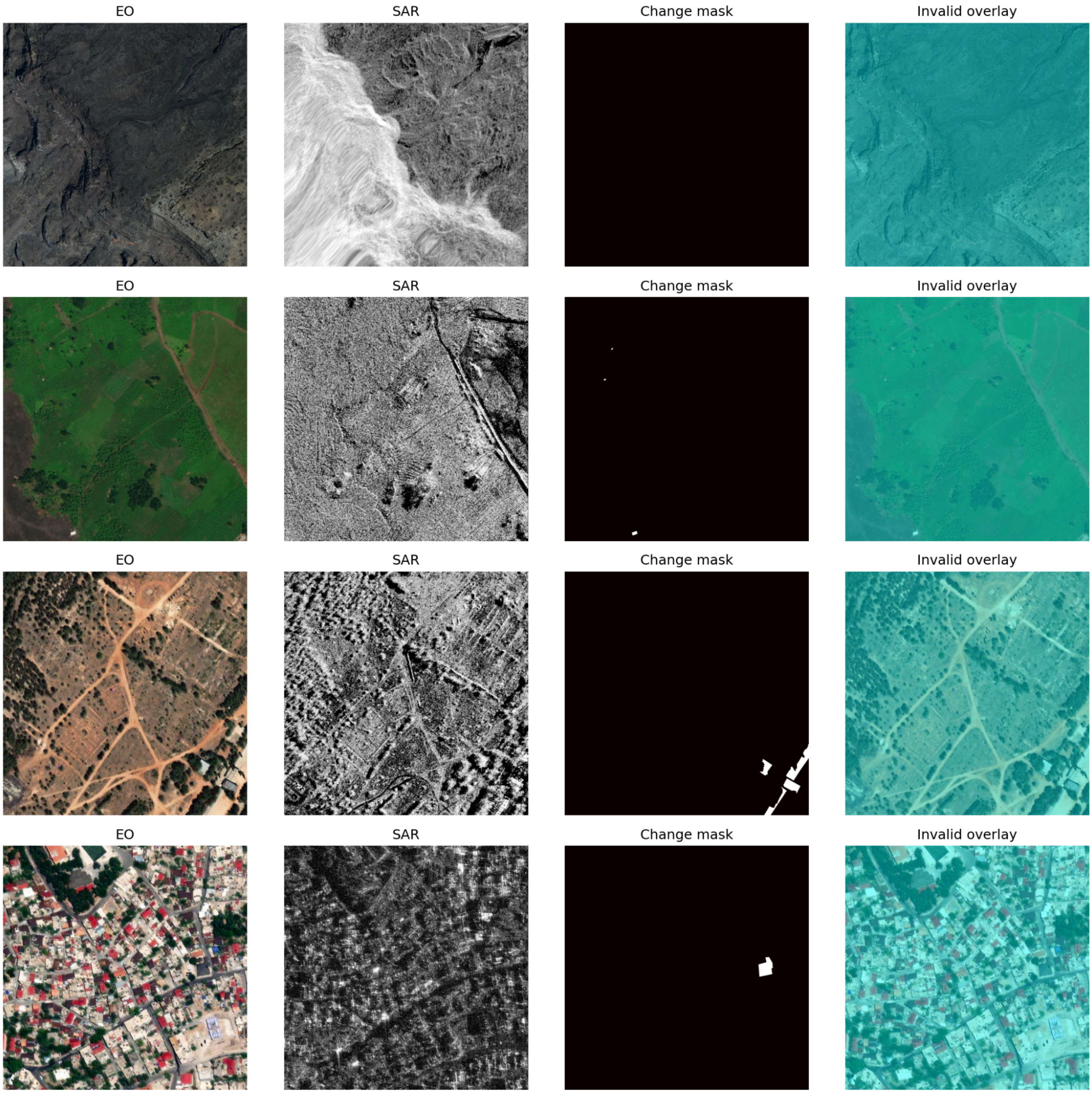


Figure 1: Example EO, SAR, and target mask samples from the training split

4.3 Class Imbalance Analysis

One of the strongest dataset characteristics was severe foreground imbalance. Most pixels in the dataset belong to the no-change class, while change regions occupy only a small percentage of the total image area.

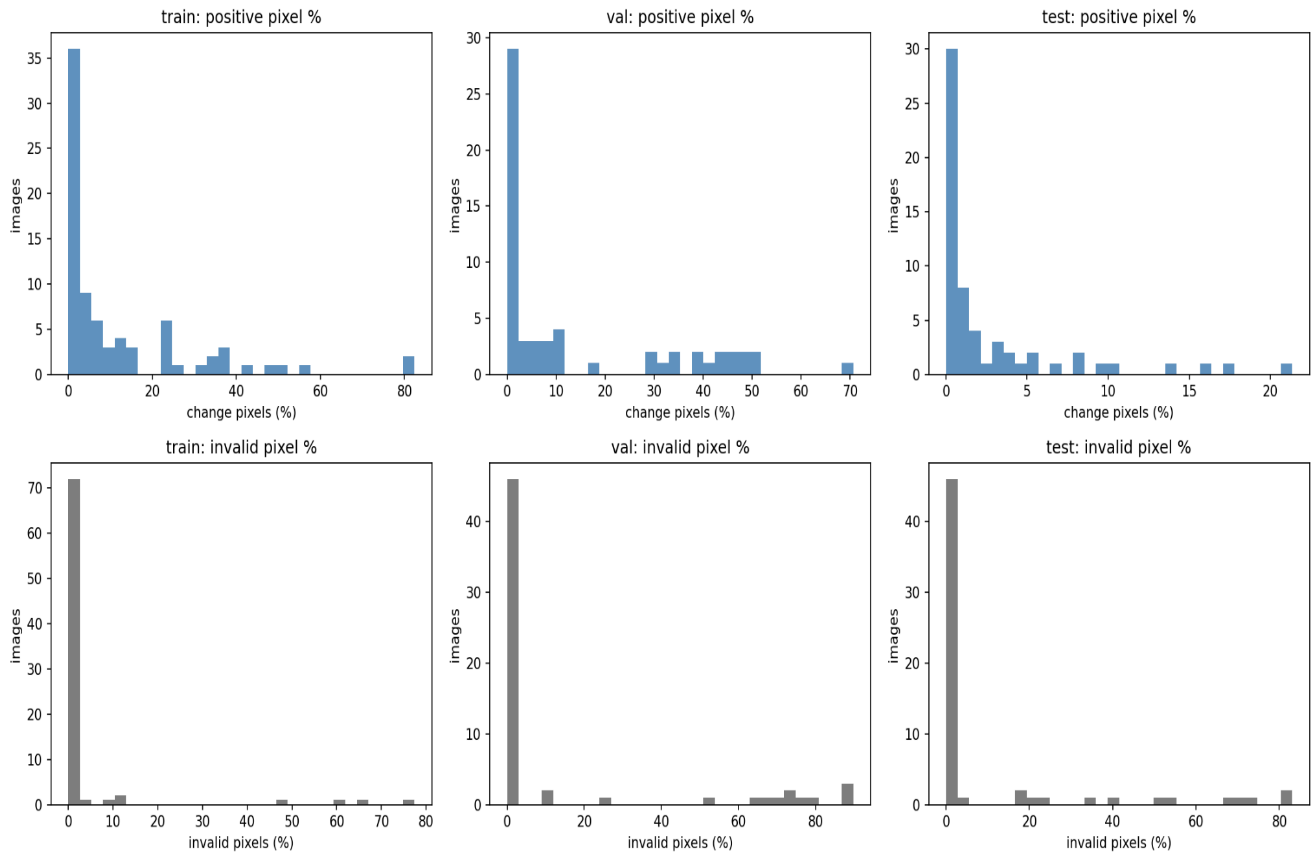


Figure 2: Distribution of positive pixel percentage and invalid region percentage across the dataset

The histogram analysis showed that many images contained extremely low positive pixel ratios. Only a relatively small number of scenes contained large damaged regions. This imbalance creates two practical problems:

1. The model can become biased toward predicting background.
2. Accuracy becomes a misleading evaluation metric because true negatives dominate the confusion matrix.

This observation motivated the use of focal loss, Dice loss, and positive-biased patch sampling during training.

4.4 Invalid and Partially Overlapping Regions

During inspection, several images were found to contain invalid regions near borders or diagonal edges. These regions likely originate from imperfect overlap or registration differences between EO and SAR acquisitions.

In some scenes, these invalid areas formed sharp artificial boundaries that visually resembled real structural changes. Early experiments showed that the model frequently produced false positives around these regions when no masking was applied.

Because of this, validity masking was introduced during:

- loss computation,
- metric computation,
- postprocessing,
- final prediction suppression.

This became an important component of the final evaluation pipeline.

4.5 SAR Intensity Distribution Analysis

SAR intensity distributions were also examined across the train, validation, and test splits.

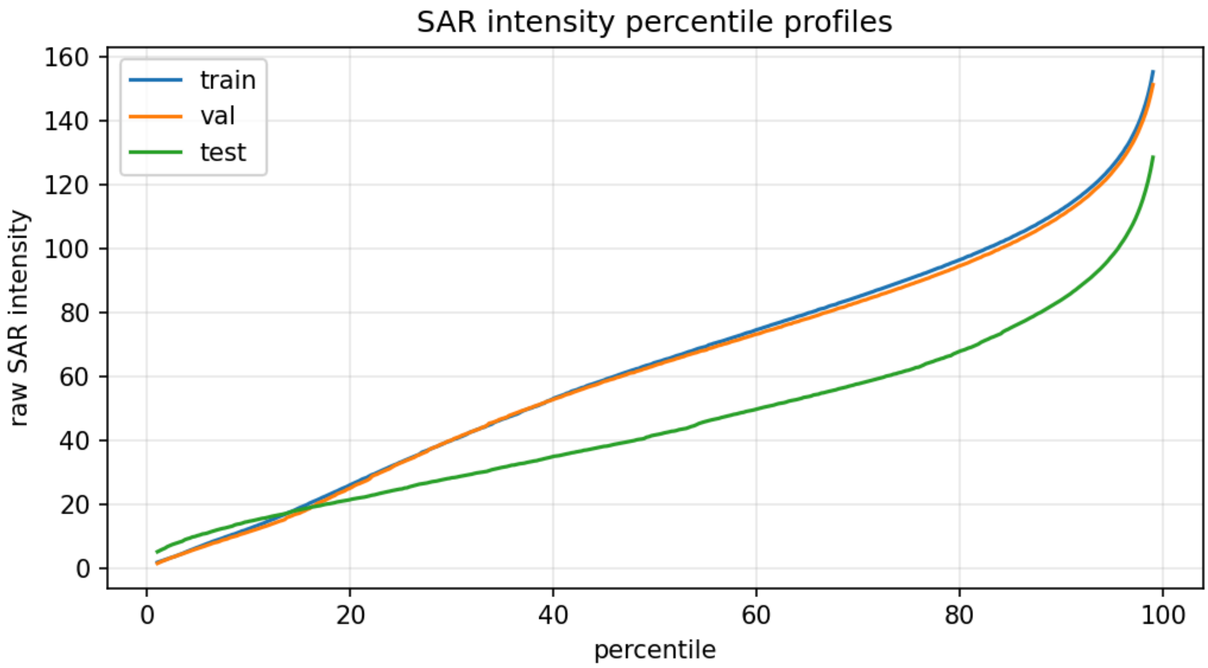


Figure 3: SAR percentile distribution comparison across dataset splits

The percentile analysis showed noticeable differences between training and test SAR distributions. In particular, the test split exhibited lower intensity statistics compared to the training data. This indicated a moderate train-test distribution shift.

This observation became important later during inference and threshold selection. Models trained directly on the training distribution tended to produce unstable predictions on certain test scenes, particularly in noisy urban regions. Per-image SAR normalization was later introduced to partially reduce this effect.

4.6 Key Observations from EDA

The exploratory analysis phase highlighted several practical challenges that shaped the remainder of the implementation:

- severe class imbalance,
- sparse and localized damage regions,
- strong EO-SAR modality differences,
- SAR speckle noise,
- invalid co-registration borders,
- train-test distribution variability.

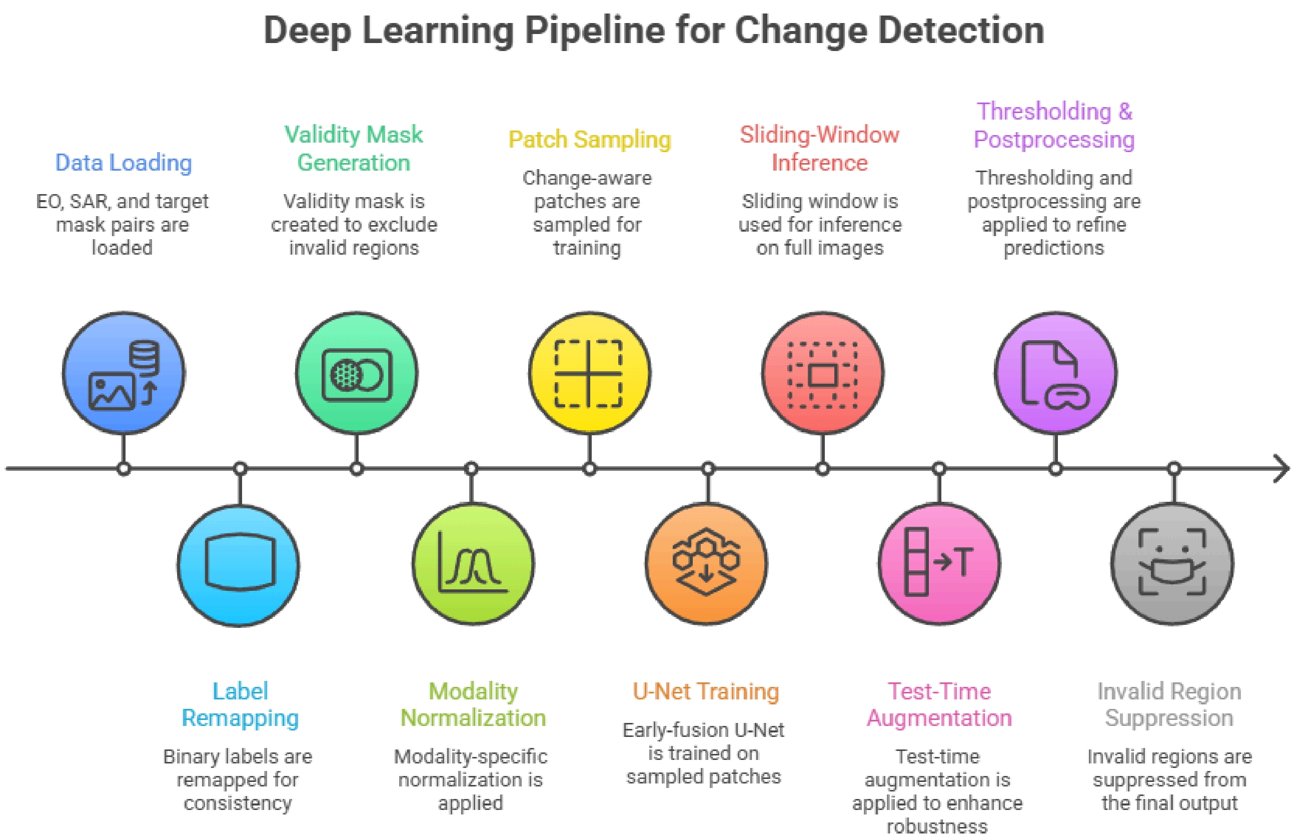
Rather than immediately increasing model complexity, the later pipeline design focused first on improving preprocessing quality, stable optimization, masking strategies, and inference refinement techniques.

5. Methodology

5.1 Overall Pipeline

The final pipeline was designed as a lightweight EO-SAR segmentation system with additional preprocessing and inference refinements to handle dataset-specific challenges. The complete workflow consists of:

1. Loading EO, SAR, and target mask pairs
2. Binary label remapping
3. Validity mask generation
4. Modality-specific normalization
5. Change-aware patch sampling
6. Early-fusion U-Net training
7. Sliding-window inference
8. Test-time augmentation (TTA)
9. Thresholding and postprocessing
10. Invalid-region suppression



The implementation intentionally prioritized stability and interpretability over architectural complexity. Instead of using large multimodal transformer models or attention-heavy fusion strategies, the focus was placed on handling class imbalance, SAR variability, and invalid regions more effectively.

5.2 Input Representation and Label Remapping

Each training sample consisted of:

- pre-event EO RGB imagery,
- post-event SAR imagery,
- pixel-level annotation mask.

Following the assignment specification, the original labels were converted into a binary setup where:

- background and intact regions were mapped to no-change,
- damaged and destroyed regions were mapped to change.

The EO RGB channels and SAR channel were concatenated directly to form the network input. This corresponds to an early-fusion setup similar to FC-EF style change detection pipelines.

The final input representation therefore contained:

- 3 EO channels,
- 1 SAR channel.

5.3 EO and SAR Preprocessing

EO and SAR imagery were normalized separately because the two modalities follow very different intensity distributions.

EO preprocessing mainly involved standard image normalization and scaling. SAR preprocessing required additional attention due to:

- large intensity variation across scenes,
- speckle noise,
- visible distribution differences between train and test splits.

Exploratory analysis showed that the test SAR distributions differed noticeably from the training split. To reduce sensitivity to these variations, per-image SAR normalization was introduced during preprocessing. This improved prediction stability compared to using a single global normalization strategy.

No explicit SAR despeckling method was applied in the final implementation, although several qualitative errors suggested that speckle noise remained an important source of false positives.

5.4 Validity Masking

Several dataset samples contained partially invalid or weakly overlapping border regions. These areas frequently produced artificial edge responses during early experiments.

To reduce their impact, validity masks were generated and applied throughout the pipeline. Invalid pixels were excluded from:

- loss computation,
- metric computation,
- final prediction evaluation.

During inference, invalid regions were also explicitly suppressed after postprocessing to avoid artificial activations near scene boundaries.

This masking step had a noticeable effect on reducing border-related false positives.

5.5 Patch Sampling Strategy

The dataset contains severe foreground imbalance, with most pixels belonging to the no-change class. Training directly on randomly sampled patches resulted in many batches containing very few positive pixels.

To address this issue, change-aware patch sampling was introduced. Approximately 70% of sampled patches were biased toward regions containing positive change pixels.

This helped improve:

- foreground exposure during training,
- gradient stability,
- recall on sparse damaged regions.

The remaining patches were sampled normally to maintain background diversity and reduce overfitting to positive regions.

5.6 Network Architecture

The final model uses a lightweight early-fusion U-Net architecture. EO RGB channels and SAR inputs were concatenated before entering a shared encoder-decoder segmentation network.

A relatively simple architecture was intentionally selected for several reasons:

- limited training time and compute,
- focus on data handling and evaluation quality,
- easier debugging and experimentation,
- lower risk of overfitting on limited data.

Skip connections in the U-Net decoder helped preserve spatial information required for pixel-level localization of sparse change regions.

More complex multimodal architectures and transformer-based fusion methods were reviewed during the literature survey phase but were not implemented in the final pipeline.

5.7 Loss Functions

The final training objective combined masked focal loss and Dice loss.

Focal loss was used to reduce the dominance of easy background pixels during optimization. Since the dataset is highly imbalanced, standard binary cross-entropy often biased predictions toward the no-change class.

Dice loss was added to directly optimize overlap quality between predictions and ground truth masks. This was particularly useful for sparse foreground regions where IoU and F1-score are more informative than pixel accuracy.

Both losses were computed only on valid pixels using the generated validity masks.

5.8 Data Augmentation

Basic geometric augmentations were applied during training to improve spatial variability and reduce overfitting. These augmentations included operations such as:

- horizontal flips,
- vertical flips,
- spatial transformations.

The augmentation pipeline was intentionally kept lightweight to avoid introducing unrealistic multimodal distortions between EO and SAR inputs.

5.9 Inference Pipeline

Inference was performed using a sliding-window strategy rather than full-image prediction. This reduced memory usage and allowed higher-resolution evaluation.

To reduce visible seam artifacts between overlapping windows, Gaussian-weighted blending was applied during reconstruction.

Geometric test-time augmentation (TTA) was also used during inference. Predictions from augmented views were averaged before thresholding. This produced slightly more stable outputs, particularly near fragmented change regions.

5.10 Thresholding and Postprocessing

The segmentation model outputs probability maps that require thresholding to generate final binary masks.

Threshold selection was treated as an explicit evaluation step rather than assuming a default threshold of 0.5. Validation experiments showed that lower thresholds increased recall substantially but also introduced excessive false positives. The final threshold was selected based on validation F1-score behavior.

After thresholding, connected-component filtering was applied to remove very small isolated predictions smaller than 24 pixels.

This helped suppress:

- speckle-induced activations,
- isolated noisy detections,
- tiny edge artifacts.

Finally, invalid regions were removed from the predictions before computing evaluation metrics.

6. Training Setup and Evaluation Metrics

6.1 Training Configuration

The model was trained using the predefined train and validation splits provided with the assignment. Training was performed using patch-based mini-batches due to memory constraints and the large spatial dimensions of the original scenes.

The implementation was developed and trained primarily in a GPU-based notebook environment. Mixed precision training was used where possible to reduce memory usage and improve training speed.

Table 1: Training Configuration

Parameter	Value
Architecture	Lightweight Early-Fusion U-Net
Input	EO RGB + SAR
Image Size	256 × 256
Loss Function	Masked Focal + Dice
Optimizer	AdamW
Learning Rate	1e-4
Batch Size	8
Epochs	20
Inference	Sliding window + geometric TTA
Postprocessing	Connected-component filtering
Hardware	Kaggle GPU environment
Runtime	~4–5 hours total training and evaluation

6.2 Validation Strategy

The validation split was used throughout development for:

- monitoring convergence,
- threshold selection,
- postprocessing analysis,
- qualitative inspection,
- model comparison.

The final threshold was selected using validation F1-score behavior rather than using a fixed default threshold. This became important because prediction calibration varied noticeably across scenes due to SAR intensity differences and class imbalance.

Validation metrics were consistently stronger than test metrics during experimentation, suggesting moderate train-test distribution shift.

6.3 Evaluation Metrics

The assignment required reporting:

- Intersection over Union (IoU),
- Precision,
- Recall,
- F1-score.

All metrics were computed for the positive change class.

Since the dataset contains severe foreground imbalance, additional metrics and analysis were included to better characterize model behavior.

6.4 Why Accuracy Alone Is Misleading

Pixel accuracy is not a reliable metric for this dataset because unchanged regions dominate most scenes. A model can achieve high accuracy simply by predicting nearly all pixels as background.

This behavior was observed during experimentation. Even when IoU and F1-score remained relatively modest, overall accuracy stayed above 90% because the number of true negative pixels was extremely large compared to the number of positive pixels.

For this reason, IoU, F1-score, precision-recall analysis, and confusion matrix inspection were treated as more informative indicators of actual segmentation quality.

6.5 Matthews Correlation Coefficient (MCC)

Matthews Correlation Coefficient (MCC) was additionally included because it provides a more balanced evaluation under heavy class imbalance.

Unlike accuracy, MCC considers:

- true positives,
- true negatives,
- false positives,
- false negatives

simultaneously in a single metric. This makes it more reliable when the negative class dominates the dataset.

The validation MCC was substantially higher than the final test MCC, which further reflected the observed generalization gap between splits.

6.6 Precision-Recall Tradeoff

The model exhibited a strong precision-recall tradeoff during threshold experiments.

Lower thresholds produced:

- higher recall,
- better foreground coverage,
- but significantly more false positives.

Higher thresholds reduced noisy activations but also removed many sparse damaged regions.

Precision-recall behavior is discussed further in the Threshold Analysis section.

6.7 Confusion Matrix and Error Profile

Confusion matrix analysis was used to better understand the types of prediction errors produced by the model.

The final predictions showed:

- a large number of true negatives,
- moderate recall,
- relatively high false positive counts.

Many false positives originated from:

- SAR texture responses,
- urban edges,
- noisy border regions,
- fragmented speckle patterns.

This error profile is discussed further in the qualitative and failure analysis sections.

7. Experimental Results

7.1 Quantitative Performance

The final pipeline was evaluated on the provided validation and test splits using IoU, F1-score, Precision, Recall, and MCC. Metrics were computed after thresholding, connected-component filtering, and invalid-region suppression.

The final test performance is summarized below.

Split	Validation	Test
Stage	Threshold Selection	Final evaluation
IoU	0.491075304445814	0.141667049295497
F1/Dice	0.658686116350093	0.248175765530289
Precision	0.616622112481325	0.173512776900224
Recall	0.706909231676447	0.435627323758837
MCC	0.609191939672205	0.23997931956785
AP	0.725552650678507	-

Table 2: Final Validation and Test Metrics

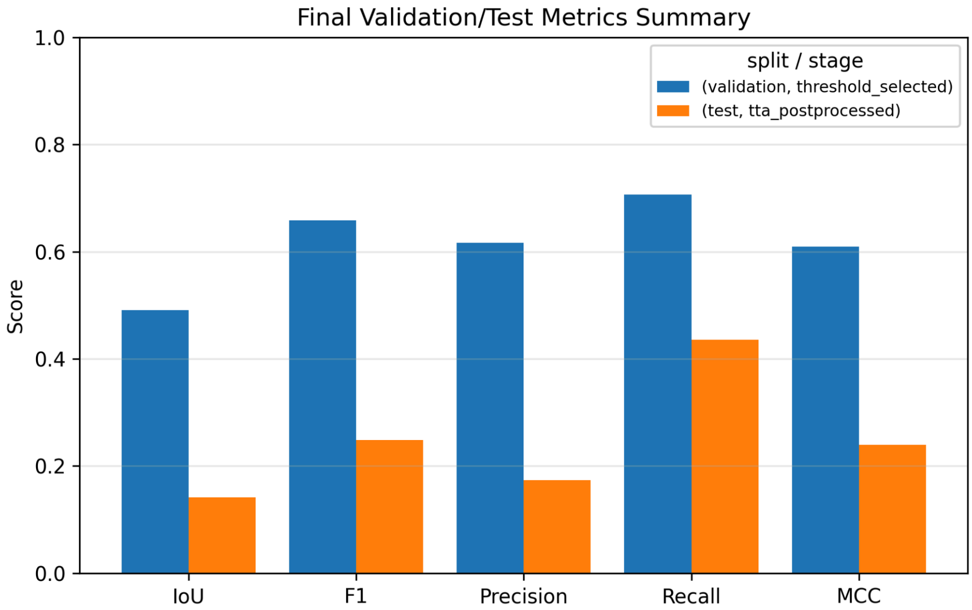


Figure 4: Validation and Test Metric Summary

The final test split achieved:

- IoU: 0.1417
- F1-score: 0.2482
- Precision: 0.1735
- Recall: 0.4356
- MCC: 0.2400

The results indicate that the model was able to recover a reasonable portion of positive regions, but at the cost of a relatively large number of false positives. Recall remained substantially higher than precision, which suggests that the model favored foreground detection sensitivity over conservative predictions.

7.2 Validation vs Test Performance

One consistent observation during experimentation was the noticeable gap between validation and test performance.

Validation metrics remained considerably stronger than the final test metrics, particularly for:

- F1-score,
- MCC,
- average precision.

The validation Average Precision (AP) reached approximately 0.7256, while the final test MCC dropped to 0.2400.

This behavior suggests moderate train-test distribution shift, which was also visible during exploratory SAR intensity analysis. The test SAR distributions differed from the training split, making calibration and threshold selection less stable on unseen scenes.

The performance gap also indicates that the model may have partially adapted to validation-specific characteristics despite maintaining relatively lightweight architecture complexity.

7.4 Confusion Matrix Analysis

The confusion matrix provides additional insight into the prediction behavior of the final pipeline.

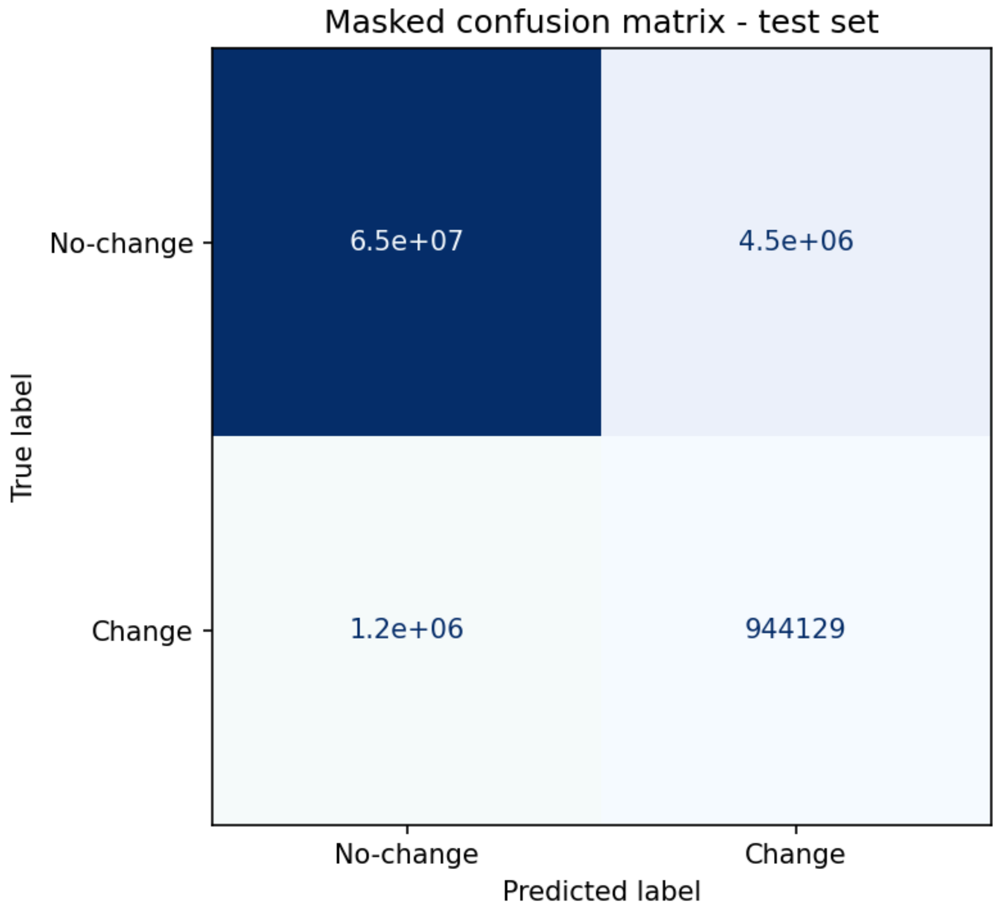


Figure 5: Confusion Matrix after Masking and Postprocessing

The final test predictions produced:

- TP: 944,129
- FP: 4,497,136
- FN: 1,223,157
- TN: 65,332,616

The very large number of true negatives explains why overall accuracy remained high despite relatively modest IoU and F1-score values. This further reinforces why accuracy alone is not a meaningful metric for this dataset.

The confusion matrix also shows that false positives remain the dominant source of error. Many of these activations appeared around:

- textured SAR regions,
- urban structures,
- roads,
- invalid overlap boundaries.

These behaviors are explored further in the qualitative analysis and failure analysis sections.

7.5 Effect of Postprocessing

Postprocessing contributed noticeably to prediction cleanup, particularly in reducing tiny isolated activations produced by SAR speckle noise.

Connected-component filtering removed many fragmented predictions smaller than 24 pixels, while invalid-region suppression reduced artificial detections near scene borders.

Qualitatively, these refinements improved:

- visual consistency,
- boundary cleanliness,
- stability of final masks.

However, postprocessing also introduced tradeoffs. Some very small true positive regions were removed together with noisy detections, which likely contributed to reduced recall in a subset of sparse scenes.

8. Threshold Analysis

8.1 Motivation for Threshold Tuning

The segmentation model produces continuous probability maps rather than direct binary predictions. Converting these probabilities into final change masks requires selecting an appropriate decision threshold.

For balanced segmentation tasks, a default threshold of 0.5 is often sufficient. However, in this dataset the combination of:

- severe foreground imbalance,
- SAR noise,
- train-test variability,
- sparse positive regions

made threshold selection considerably more important.

During early experiments, small threshold changes produced large shifts in precision and recall behavior. Because of this, threshold tuning was treated as an explicit evaluation step rather than a fixed postprocessing choice.

8.2 Validation Precision-Recall Analysis

Threshold selection was performed using the validation split by analyzing precision-recall behavior across multiple operating points.

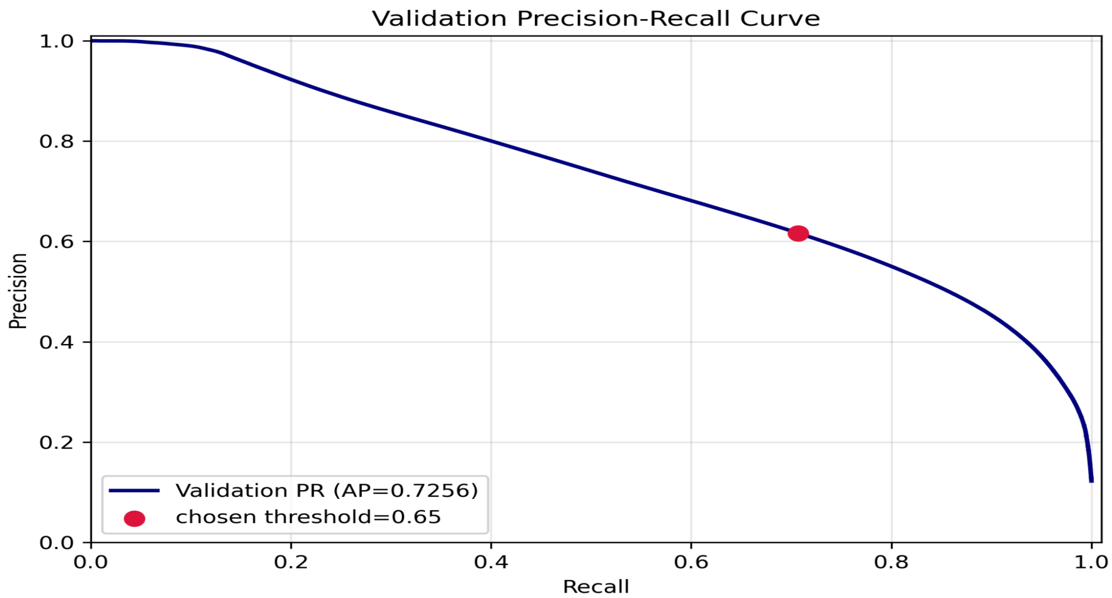


Figure 6: Validation Precision-Recall Curve

The validation curve showed that lower thresholds increased recall substantially, but also introduced a large number of false positives. Many of these false activations appeared in:

- noisy SAR regions,
- urban textures,
- partially invalid borders,
- fragmented edge structures.

Higher thresholds produced visually cleaner masks and improved precision, but also removed many sparse positive regions that occupied only a few pixels.

The final threshold of approximately 0.70 was selected because it provided a more stable balance between foreground coverage and prediction noise.

8.3 Selected Operating Point

The final threshold was selected using validation-set precision-recall analysis together with qualitative inspection of prediction stability. A threshold of approximately 0.70 provided a reasonable balance between foreground coverage and excessive false activations.

Lower thresholds increased recall but produced noisy fragmented predictions, while higher thresholds suppressed many sparse positive regions. The selected operating point was therefore chosen as a practical compromise rather than an optimal threshold for every scene.

8.4 Interaction with Postprocessing

Thresholding and postprocessing were evaluated together during validation. Connected-component filtering reduced many isolated noisy activations, particularly in textured SAR regions, but occasionally removed very small valid detections. The final configuration was selected using both quantitative metrics and qualitative inspection across multiple scenes.

8.5 Observations

Threshold sensitivity varied across scenes, particularly under SAR intensity variability and sparse foreground conditions. This suggests that adaptive thresholding or calibration-based approaches may improve generalization in future work.

9. Qualitative Analysis

9.1 Overview

Quantitative metrics provide a useful summary of overall performance, but they do not fully capture the spatial behavior of the model. Qualitative inspection was therefore used extensively throughout development to better understand:

- prediction consistency,
- false positive patterns,
- boundary behavior,
- sparse region detection,
- effects of preprocessing and postprocessing.

Representative prediction examples are shown below.

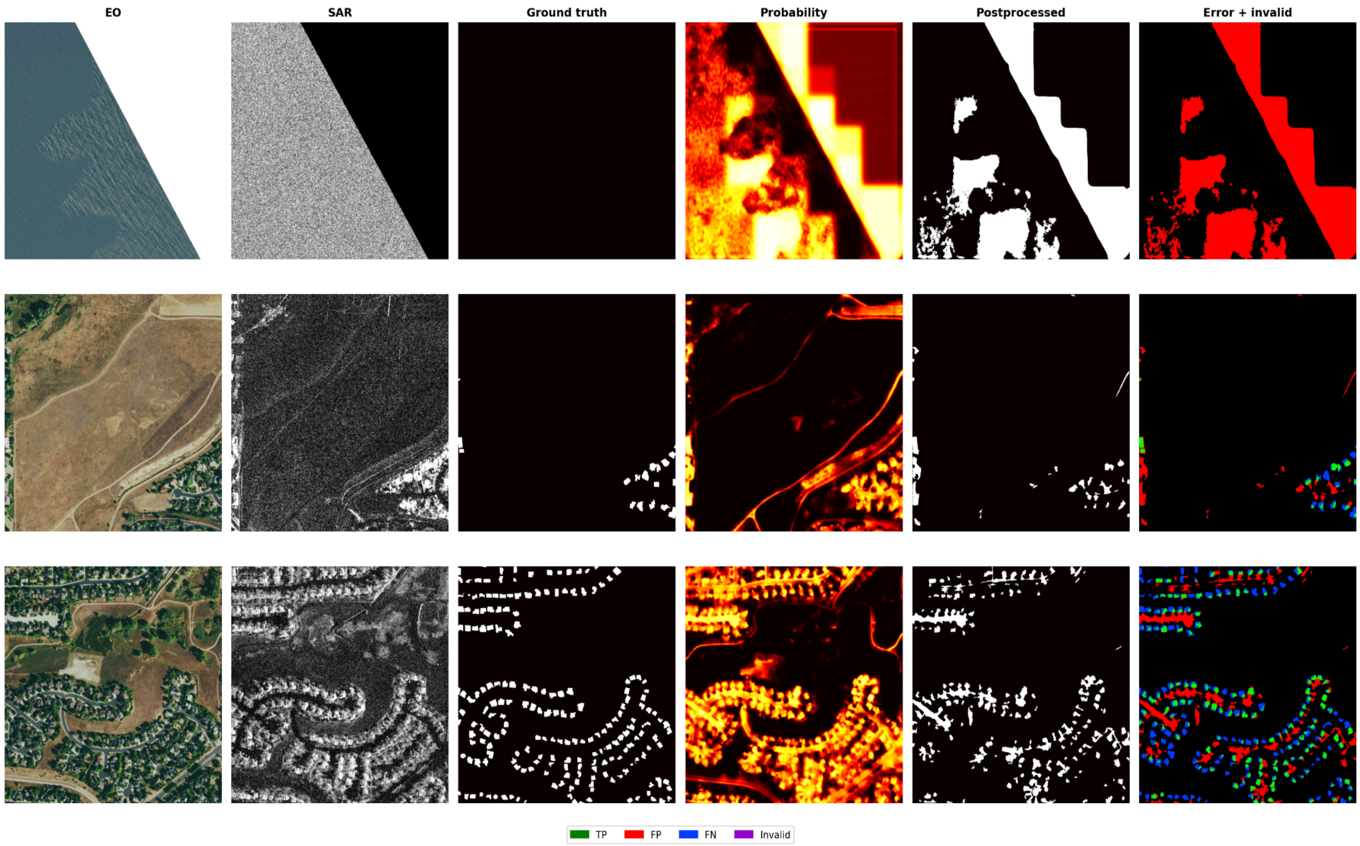


Figure 7: Qualitative prediction examples with validity masking

9.2 Successful Prediction Cases

Several scenes showed reasonably accurate localization of damaged regions despite the challenging EO-SAR modality gap.

The model generally performed better when:

- damaged regions were spatially contiguous,
- SAR contrast was relatively stable,
- invalid border regions were limited,
- foreground structures occupied larger areas.

In these cases, the predicted masks were able to recover major damaged regions while maintaining acceptable spatial alignment with the ground truth masks.

The combination of:

- positive-biased sampling,
- focal + Dice loss,
- threshold tuning,
- connected-component filtering

helped improve foreground continuity and reduce fragmented activations in cleaner scenes.

Postprocessing was particularly useful in removing isolated speckle-induced detections that appeared in earlier inference outputs.

9.3 Effect of Validity Masking

Validity masking had a noticeable impact on qualitative prediction quality, especially near scene boundaries.

Without masking, the model frequently produced strong activations near:

- diagonal overlap edges,
- partially invalid regions,
- abrupt SAR intensity transitions.

Suppressing invalid regions during both evaluation and final prediction refinement reduced many of these artificial detections.

This improvement was more visually noticeable than what aggregate metrics alone suggested.

9.4 General Prediction Behavior

Prediction quality varied across scenes depending on SAR contrast, scene complexity, and foreground sparsity. Larger contiguous damaged regions were generally detected more reliably than very small fragmented structures. Postprocessing and validity masking improved overall visual consistency by reducing isolated activations and unstable boundary responses.

Detailed failure modes are discussed further in Section 10.

9.5 Small and Sparse Object Behavior

The model showed mixed performance on very small positive regions. Larger damaged areas were usually detected more consistently, while tiny fragmented structures were sometimes weakly activated or removed during postprocessing.

10. Failure Analysis

10.1 Overview

Failure analysis was performed to better understand the dominant error modes of the final pipeline beyond aggregate metrics. While the model was able to recover many positive regions, several consistent failure patterns appeared across validation and test scenes.

The most common issues were:

- false positives caused by SAR texture,
- activations near invalid borders,
- fragmented predictions,
- missed small damaged regions,
- unstable behavior under distribution shift.

Representative failure cases are discussed below.

10.2 Invalid Border and Co-registration Failures

One of the strongest recurring failure modes appeared near partially invalid or weakly aligned image borders.

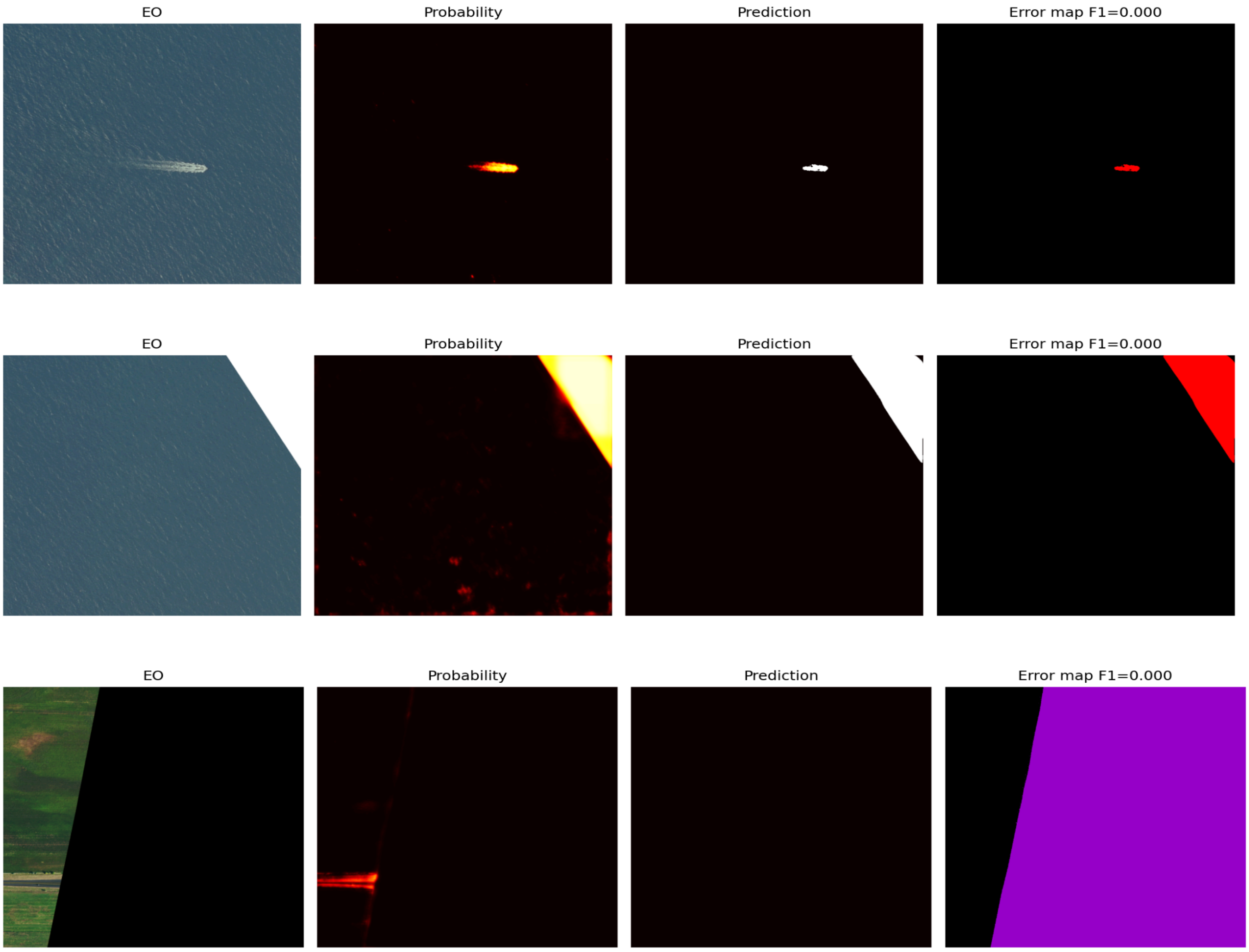


Figure 8: Failure examples near invalid overlap regions

In several scenes, diagonal or irregular overlap boundaries produced sharp transitions in SAR intensity. These artificial edges often resembled structural changes and triggered strong activations from the segmentation model.

Although validity masking reduced many of these artifacts, some residual false positives still remained near boundary regions. This suggests that masking alone cannot fully compensate for imperfect cross-modal alignment.

These errors became more severe when:

- the invalid region occupied a large area,
- SAR contrast changed abruptly,
- urban structures were located near borders.

10.3 SAR Texture and Speckle-Induced False Positives

SAR speckle noise was another major source of prediction instability.

The model frequently activated on:

- textured urban regions,
- roads,
- bright scattering structures,
- noisy high-frequency SAR patterns.

In many cases, these activations were visually plausible because SAR textures can resemble fragmented structural damage. However, they did not correspond to actual positive labels in the target masks.

This issue was especially visible in dense urban scenes where:

- buildings generated strong backscatter variation,
- roads created linear responses,
- local SAR intensity fluctuations became highly irregular.

Connected-component filtering reduced many isolated detections, but larger textured regions still produced persistent false positives.

10.4 Missed Sparse Positive Regions

The model also struggled with very small or weakly visible damage regions.

Typical failure patterns included:

- partial detection of fragmented structures,
- disconnected predictions,
- weak activation on small positives,
- complete suppression after thresholding.

This behavior is closely related to the dataset imbalance problem. Since foreground regions occupy only a very small fraction of the image area, the optimization process naturally becomes biased toward conservative background predictions.

The postprocessing pipeline further amplified this tradeoff. Small connected components were intentionally removed to suppress noise, but some valid detections were also removed together with false positives.

10.5 Validation-Test Generalization Gap

Another important observation was the noticeable gap between validation and test performance.

Validation metrics remained substantially stronger than the final test results despite using the same overall pipeline. Several factors likely contributed to this:

- SAR intensity distribution differences,
- geographic variability,
- scene diversity,
- different foreground sparsity patterns.

The SAR percentile analysis performed during EDA also suggested moderate train-test shift, particularly in intensity statistics.

This affected:

- prediction calibration,
- threshold stability,
- probability confidence,
- foreground separation quality.

The issue became more noticeable in scenes with:

- weaker SAR contrast,
- dense urban clutter,
- heavy texture variability.

10.6 Limitations of the Lightweight Fusion Strategy

The early-fusion U-Net approach was computationally efficient and relatively stable to train, but it also introduced limitations.

Since EO and SAR channels were concatenated directly at the input stage, the network had limited ability to explicitly model:

- modality-specific representations,
- cross-modal feature relationships,
- attention-based alignment.

More advanced multimodal fusion architectures may improve robustness under severe modality differences, although they would also increase training complexity and computational cost.

The current implementation therefore represents a tradeoff between:

- simplicity,
- training stability,
- interpretability,
- and multimodal modeling capacity.

10.7 Summary of Failure Modes

The main failure patterns observed throughout experimentation can be summarized as follows:

Failure Type	Primary Cause
Border false positives	Invalid overlap regions and alignment inconsistencies
Urban texture activations	SAR speckle and high-frequency backscatter
Missed small positives	Severe class imbalance and thresholding
Fragmented masks	Sparse targets and noisy predictions
Validation-test performance gap	SAR distribution shift and scene variability

These observations motivated many of the preprocessing and postprocessing choices used in the final pipeline and also highlight several areas for future improvement.

11. Limitations and Future Work

11.1 Current Limitations

Although the final pipeline produced reasonable qualitative predictions under limited compute and development constraints, several limitations remain.

The largest issue was the high false positive rate, particularly in noisy SAR regions and near partially invalid borders. While preprocessing and postprocessing reduced many isolated activations, the model still struggled to reliably distinguish between true structural changes and complex SAR texture patterns.

Another limitation was the noticeable validation-test performance gap. The exploratory analysis suggested moderate train-test distribution shift in SAR intensity statistics, and the final results indicated that the model did not generalize equally well across all scenes. Prediction calibration and threshold stability varied considerably depending on the scene characteristics.

The lightweight early-fusion design also imposed representational limitations. Concatenating EO and SAR inputs directly at the encoder stage simplifies training, but it does not explicitly model modality-specific feature interactions. As a result, the network may struggle to fully capture the semantic relationship between EO appearance information and SAR backscatter behavior.

The current pipeline also relies on several heuristic components, including:

- threshold tuning,
- connected-component filtering,
- invalid-region suppression.

While these refinements improved overall prediction quality, they are not learned directly by the model and may not generalize optimally across all scenes.

Finally, no explicit SAR despeckling or domain adaptation strategy was implemented. Speckle noise and SAR distribution variability remained major sources of instability throughout experimentation.

11.2 Potential Improvements

Several extensions could improve the current pipeline in future iterations.

One natural direction would be to explore dual-stream multimodal architectures where EO and SAR inputs are processed separately before feature fusion. This may help the network learn modality-specific representations more effectively than direct early fusion.

Attention-based fusion or transformer-based approaches could also improve cross-modal interaction modeling, particularly in scenes with weak semantic correspondence between EO and SAR imagery. However, these methods would require substantially more compute and careful regularization.

Additional work could also focus on improving SAR preprocessing. Possible directions include:

- explicit despeckling methods,
- log-intensity transformations,
- adaptive normalization strategies,
- scene-wise calibration techniques.

The handling of class imbalance could likely be improved further using:

- hard negative mining,
- focal-Tversky style losses,
- boundary-aware objectives,
- uncertainty-weighted sampling.

Another possible improvement would be adaptive thresholding instead of using a single global threshold across all scenes. Since prediction calibration varied noticeably across different SAR distributions, scene-aware threshold selection may improve generalization.

Finally, larger-scale experimentation with:

- stronger augmentation pipelines,
- longer training schedules,
- larger encoder backbones,
- multi-scale context aggregation

could improve segmentation quality, particularly for fragmented and sparse damage regions.

11.3 Practical Takeaways

A major takeaway from this work is that EO-SAR change detection performance depends heavily on preprocessing quality and dataset understanding, not only on model complexity.

Several improvements in the final pipeline came from:

- validity masking,
- SAR normalization,
- threshold tuning,
- postprocessing refinement,

- sampling strategy adjustments

rather than architectural changes alone.

The experiments also highlighted how strongly SAR noise, sparse positives, and train-test variability influence real-world segmentation behavior. Even relatively lightweight models can produce reasonable outputs when these practical issues are handled carefully, although substantial room for improvement still remains.

12. Conclusion

This work presented a lightweight EO-SAR binary change detection pipeline for identifying damaged regions from paired pre-event EO imagery and post-event SAR imagery. The experiments highlighted several practical challenges associated with multimodal remote sensing data, particularly severe class imbalance, SAR variability, and partially invalid overlap regions.

The final pipeline combined early-fusion segmentation with validity masking, SAR normalization, change-aware sampling, and postprocessing refinements. These components improved prediction stability and reduced many noisy activations without requiring large multimodal architectures.

The main limitation of the current approach remains the high false positive rate in difficult SAR regions together with reduced generalization under train-test distribution shift. Despite these limitations, the project demonstrates that careful preprocessing, evaluation, and inference refinement can substantially improve EO-SAR change detection performance even with relatively lightweight models.

13. References

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4. Shermeyer, J., Hogan, D., Brown, J., et al., "SpaceNet 6: Multi-Sensor All Weather Mapping Dataset," *CVPR Workshops*, 2020.
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Appendix A. Additional Qualitative Results

Additional qualitative examples are provided to illustrate prediction behavior across different scene types. The examples include EO imagery, SAR imagery, ground-truth masks, probability maps, final postprocessed predictions, and error overlays.

The visualizations show that the model was generally able to localize larger changed regions, while smaller fragmented regions and boundary areas remained more challenging.

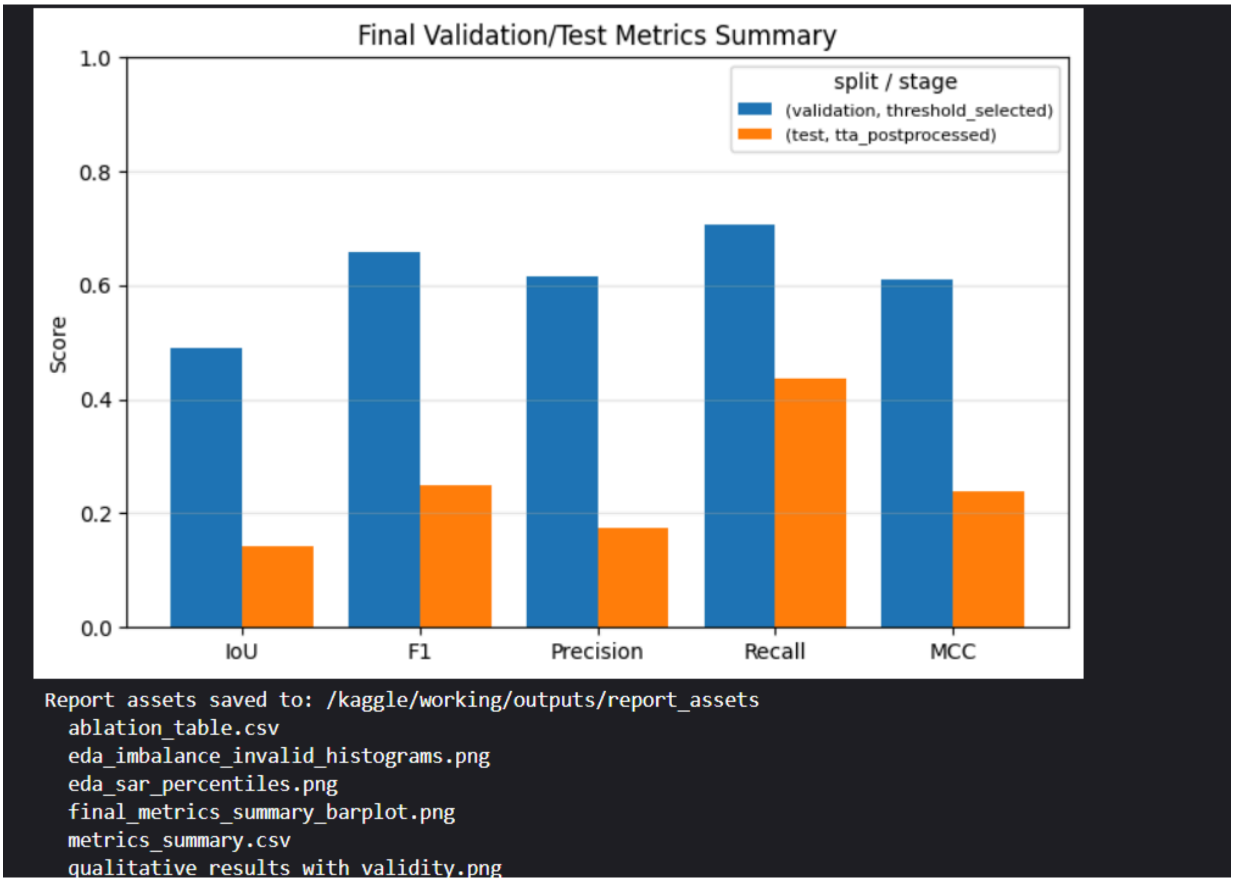

Figure A1: Additional qualitative prediction examples from the test split.

Appendix B. Threshold Calibration and Postprocessing Summary

Post-training calibration was performed on the validation split to select the final threshold and connected-component filtering configuration. Different threshold and morphology settings were evaluated using validation F1 score together with qualitative inspection of prediction stability.

The final configuration used:

- threshold = 0.70
- minimum component size = 64 pixels
- no morphological opening

This setting provided the best balance between recall and excessive false activations.

Threshold	Min Component Size	Morphology	F1	Precision	Recall
0.60	64	opening_3×3	0.2376	0.1479	0.6043
0.65	64	none	0.2477	0.1615	0.5311
0.70	64	none	**0.2482**	**0.1735**	0.4356
0.75	64	none	0.2163	0.1772	0.2777

Table B1: Summary of selected threshold calibration results.

Appendix C. Runtime and Resource Log

C.1 Development Environment

Item	Details
Development Platform	Kaggle Notebook Environment
Framework	PyTorch
GPU Used	Kaggle GPU T4×2
VRAM	16 GB
Inference Strategy	Sliding-window inference with TTA
Mixed Precision	Enabled where supported

C.2 Approximate Runtime Summary

Activity	Approximate Runtime
Dataset inspection and EDA generation	~20-30 min
Training and validation experiments	~2-2.5 hrs
Threshold calibration and postprocessing sweeps	~20-30 min
Final test inference with TTA	~10-15 min
Qualitative visualization export	~10 min

C.3 Runtime Notes

Most experiments were conducted in a Kaggle notebook environment using a lightweight early-fusion U-Net pipeline. The implementation prioritized faster experimentation cycles and lower GPU usage rather than large multimodal architectures. The training pipeline was executed with an average of 9.5 minutes per epoch.

Sliding-window inference with test-time augmentation increased inference time moderately during final evaluation.